Def. A set $B \subseteq \mathbb{R}^d$ is called a Box if:

$B = I_1 \times \cdots \times I_d$ where $I_i \subseteq \mathbb{R}$ are bounded intervals.

And, $|B| = \ell(I_1) \times \cdots \times \ell(I_d)$ is called a volumn of B.

A set $E \subseteq \mathbb{R}^d$ is called an elementary set if:

$E = \bigcup_{n=1}^{N} B_n$ where $B_1, \cdots, B_N$ are boxes.

Denote $\mathcal{E} := \{E \subseteq \mathbb{R}^d \mid E \text{ is elementary}\}$.

Lemma. Let $A, B \subset \mathbb{R}^d$ be boxes. Then,

$A \cup B$, $A \setminus B$, $A \triangle B$ are finite union of disjoint boxes.

Proof. Denote that

$A = A_1 \times \cdots \times A_d$ and $B = B_1 \times \cdots \times B_d$ where A_i, B_i are intervals.

First, $A \cap B$ is also a box, because

$$A \cap B = \left(\prod_{i=1}^{d} A_i\right) \cap \left(\prod_{i=1}^{d} B_i\right) = \prod_{n=1}^{d} (A_n \cap B_n)$$

And, for each $1 \le i \le d$, decompose $A_i = L_i \cup C_i \cup U_i$ satisfying

$L_i =$

Now, $A \setminus B = \bigcup_{J_i \in \{L_i, C_i, U_i\}} J_1 \times \cdots \times J_d$

at least one of J_i is C_i.

Fundamental theorem of elementary set.

Thm. Let $E \subset \mathbb{R}^d$ be an elementary set. Then,

E can be expressed as finite union of disjoint boxes.

Proof. Let E be an elementary set of $\mathbb{R}^d$. That is,

$$E = \bigcup_{n=1}^{N} B_n \text{ for some boxes } B_1, \cdots, B_N.$$

Using induction for N. The case of $N=1$ is trivial.

Suppose that the statement holds for $N-1$ boxes. So that the union of $B_1, \cdots, B_{N-1}$ can be expressed by union of disjoint boxes

$$\bigcup_{n=1}^{N-1} B_n = \bigsqcup_{k=1}^{M} C_k \quad \text{where } C_k \text{ is a box.}$$

And, notice that

$$E = \left(\bigcup_{n=1}^{N-1} B_n\right) \cup B_N = \left(\bigsqcup_{k=1}^{M} C_k\right) \cup B_N = \left(\bigsqcup_{k=1}^{M} C_k\right) \sqcup \left(B_N \setminus \left(\bigsqcup_{k=1}^{M} C_k\right)\right)$$

By the Lemma, a set difference of boxes is finite union of disjoint boxes

$$B_N \setminus \left(\bigsqcup_{k=1}^{M} C_k\right) =$$